Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	**Array Five**
Selected Domain	**Domain 2**
Selected SDG Track (1–17)	**SDG-8: Decent Work and Economic Growth**
Project / Solution Name	**SCDC Recovery**
Name (Leader)	**Yosef Kristian Tirta**
Email & Phone (Leader)	**MBT2309423@xmu.edu.my & +601126618368**
University Affiliation(s)	**Xiamen University Malaysia**
Number of Team Members	**5 Members**

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small and medium businesses depend on a chain of suppliers, staff and resources to deliver what they've promised their customers. When one link breaks, such as a supplier delivering late or sending fewer items than ordered, the effects spread quickly. One late delivery can leave several customer commitments at risk at once.

Large companies have supply-chain software and planning teams while most small businesses don't. They find out about disruptions through informal channels like WhatsApp messages, then work out the impact by hand: which jobs need the missing items, whether spare stock exists, which alternative suppliers can deliver in time, and at what cost. Under time pressure this is stressful and easy to get wrong. A supplier that looks cheaper may arrive too late, or a "backup" supplier may not actually have stock available.

Consider a Kuala Lumpur event planner. A supplier announces the day before two events that only 20 of 40 chairs will arrive on time. The owner must quickly decide how to protect a wedding and a corporate seminar, neither of which can be moved.

The consequences are economic and social. Missed commitments mean lost revenue, refunds, damaged reputation and lost repeat customers, and these hurt small businesses most because they operate on thin margins and rely on trust and word of mouth.

Generic AI chatbots can suggest ideas, but they can't verify whether a plan is actually feasible. Businesses need fast, reliable recovery options they can trust and stay in control of.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SCDC Recovery is a disruption-recovery assistant for small businesses. It turns a supplier problem into checked, explainable recovery options that the owner approves.
How it works:

- Understand the disruption. The owner pastes a supplier message, such as an informal WhatsApp text, or fills in a short form. A generative AI model extracts the key facts (supplier, item, quantity, new delivery date) into a structured format, and rule-based

extraction checks them. Any disagreement or uncertain value is sent back to the owner to confirm.

- Find the impact. An impact engine follows the business's dependencies (supplier → delivery → resource → job → customer commitment) and calculates which commitments are at risk and how many items are short on each day.
- Generate recovery plans. The system proposes options from real business records: alternative suppliers, spare stock, rescheduling where allowed, or asking the supplier to expedite. AI may suggest extra ideas.
- Validate every plan. A deterministic validator simulates each plan against stock, lead times, deadlines and double-booking, and labels it feasible, infeasible or needs confirmation. AI never decides feasibility, which prevents hallucinated suppliers or stock from being recommended.
- Rank and explain. Feasible plans are ranked on commitments protected, cost and certainty. AI turns the results into clear explanations of trade-offs.
- Human approval. The owner approves, edits (which triggers revalidation) or rejects a plan. The system then drafts supplier and customer messages and records every decision in an audit log.

This hybrid design uses AI where it's strongest (reading messy messages and explaining results) and rules where reliability matters.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

The React frontend sends requests over HTTPS to the FastAPI backend. The backend validates input, loads a business snapshot from the SQLite database, and runs the modules in order: AI extraction (optional) → impact analysis → recovery plan generation → deterministic validation → ranking → AI explanation. Results return to the frontend for owner approval. Every decision is saved to the database's audit log. The AI service is called only through an adapter. It never writes to the database or decides feasibility, and the system falls back to rules if the AI is unavailable.
Tech stack: React + TypeScript (Vite, Tailwind) frontend; Python FastAPI backend with SQLAlchemy and SQLite; Google Gemini 3.1 Flash-Lite through the LiteLLM adapter with a rule-based fallback; deployed on Replit (Deployments + Secrets).

Yes, Replit. Replit was used as the cloud platform to deploy the app, then develop and host the backend API for the live demo.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We followed the five design-thinking stages:

Empathise: we studied how small business owners experience disruptions: informal WhatsApp messages, time pressure and no planning tools.

Define: we framed the core need as “fast recovery options the owner can trust and stay in control of”, not just AI suggestions.

Ideate: we compared a pure-chatbot approach with a hybrid design, and chose AI for reading messages and explaining results, with deterministic rules deciding feasibility, to avoid hallucinated plans.

Prototype: we built a realistic event-planner scenario and agreed shared data contracts first, so all five members could develop modules in parallel with mock data.

Test: automated tests cover failure cases (insufficient stock, late or unknown suppliers, edited plans

becoming invalid, AI failure): 584 backend and 78 frontend tests, run with the commands in the README.

User-friendliness: clear plan statuses (works / won't work / needs confirmation), plain-language explanations, and simple approve, edit and reject actions.

Phase 2 readiness: modular, replaceable components, deployment on Replit, an AI fallback so the system runs without AI, and a planned pilot with real small businesses to refine the interface and rules.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Primary: SDG Target 8.3:** "Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises…" **Secondary: SDG Target 9.3:** "Increase the access of small-scale industrial and other enterprises… to financial services… and their integration into value chains and markets." Our tool helps micro and small businesses keep customer commitments when a supplier fails. That protects their revenue, jobs and place in supply chains, which supports their growth and resilient integration into value chains.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	MSMEs make up 96.1% of business establishments in Malaysia, contribute 39.5% of GDP and employ 48.7% of the workforce (DOSM, 2024). 55% of Malaysian businesses report that their supply chain has been impacted by geopolitical issues (UOB Business Outlook Study 2023). Yet most micro businesses have no planning tools and recover from supplier failures by hand. In our reference scenario (a supplier delivers 20 of 40 chairs late), 2 of 2 customer events (100%) on the affected day are at risk, with a 28.6% resource shortfall (50 of 70 chairs available). Without decision support, the cheapest-looking option (Delima, RM 56) arrives too late.

Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	1. Commitment protection rate: % of at-risk customer commitments covered by a validated (feasible) recovery plan. 2. Time to decide: minutes from receiving a supplier message to an approved, validated recovery plan. 3. Safety rate: % of plans shown as "feasible" that are actually infeasible. Must stay at 0%.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	By the final prototype presentation, on a test set of at least 30 synthetic disruption scenarios, plus a pilot with 3–5 Malaysian micro/small businesses if available: ● ≥ 80% of at-risk commitments protected by a validated plan, wherever a feasible option exists in the business data. ● ≥ 50% reduction in time to decision compared with the same owners solving the scenario manually (manual baseline measured in the pilot). ● 0% infeasible plans shown as feasible.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Python 3.12	Framework	Python Software Foundation License
Pydantic 2.14.1.	Framework	MIT
Pytest 9.1	Framework	MIT
SQLAlchemy	Framework	MIT
SQLite 3	Framework	Public Domain
Typescript	Framework	Apache License 2.0
Google Gemini 3.1 Flash-Lite	Model (Gemini API)	Google Gemini API Terms of Service
LiteLLM 1.100.5	Code (AI adapter library)	MIT
FastAPI	Code	MIT
Vite	Code	MIT
React	Code	MIT
Uvicorn	Code	BSD 3-Clause
React Router	Code	MIT
TanStack Query	Code	MIT
Tailwind CSS	Code	MIT
Vitest	Code	MIT
Playwright	Code	Apache 2.0

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



Yosef Kristian

Tirta

24 October 2026